CSI BISHOP SOLOMON DORAISAWMY COLLEGE OF ARTS AND SCIENCE

Team Member 1 : BOOPATHIRAJA M
Team Member 2 : MOHAMMAD SALMAN B
Team Member 3 : PRASANNA P
Team Member 4 : MIDHUN K

Platform : ServiceNow

AUTOMATING EMPLOYEE DATA

1. Introduction:

ServiceNow is a cloud-based platform used for IT service management and enterprise workflow automation. One of the common challenges in organizations is managing large volumes of employee data.Manual entry of employee records can lead to mistakes and consume significant time. Therefore, automation plays a critical role in improving efficiency.

In this project, employee data import is automated using ServiceNow Import Sets and Transform Maps. The data is first loaded into an import set table and then transformed into the target table using mapping rules.This approach helps organizations handle employee data in a structured and efficient way.

2.Problem Statement:

Many organizations still rely on manual data entry to maintain employee records. When employee information is stored in external files such as Excel sheets, employees must manually enter each record into the

system. This process requires a significant amount of time and effort.Manual data entry can also lead to several problems such as human errors, missing information, and duplicate records. When the number of employees is large, maintaining records manually becomes extremely difficult.

Another challenge is maintaining data consistency across different systems. Without automation, organizations may face difficulties in updating and synchronizing employee data.

Therefore, there is a need for an automated solution that can import employee data efficiently into the ServiceNow platform. This project aims to solve this problem by implementing an automated employee data import process using ServiceNow Import Sets and Transform Maps.

3.Objectives:

The primary objective of this project is to automate the process of importing employee data into the ServiceNow platform. Automation helps reduce manual work and improves the accuracy of data management.

Another objective of the project is to understand how ServiceNow handles data import processes. By using Import Sets, data from external files can be uploaded into the system and stored temporarily before being transferred to the final table.The project also aims to demonstrate the use of Transform Maps. Transform Maps help map the fields from the source table to the target table so that the data is stored correctly.The project further explores the concept of field mapping and customization using ServiceNow features. By implementing these features, administrators can manage employee data more efficiently.

Overall, the objective of the project is to improve productivity, reduce errors, and demonstrate the advantages of automation in data management systems.

The main objectives of this project are:

- To automate employee data import in ServiceNow
- To reduce manual data entry errors
- To understand the use of Import Sets
- To create and configure Transform Maps
- To perform field mapping using Mapping Assist
- To implement Dictionary Override
- To verify imported data in the target table

4. ABSTRACT:

Automation is an essential concept in modern organizations. Managing employee data manually can be time consuming and prone to human errors. Organizations often store employee information in external files such as Excel or CSV files. Entering this data manually into systems can lead to duplication, mistakes, and delays.

ServiceNow provides powerful features that allow organizations to automate data management processes. One such feature is Import Sets, which allows data from external files to be imported into the ServiceNow platform. Once the data is imported, Transform Maps can be used to map the fields from the source table to the target table.

This project demonstrates the automation of employee data import using ServiceNow. The project explains the steps involved in uploading data, creating transform maps, and transferring data to the final tables. It also highlights the benefits of automation such as improved efficiency, reduced errors, and better data management.

Through this project, students gain practical knowledge about ServiceNow tools and understand how automation can simplify organizational processes.

5. TOOLS USED:

The following tools and technologies were used to implement this project.

ServiceNow Platform
ServiceNow is the primary platform used to implement the project. It provides tools for workflow automation, data management, and system integration.

Excel / CSV Files
Employee data is stored in Excel or CSV files which are used as the source files for importing data into the ServiceNow system.

Web Browser
A web browser such as Google Chrome is used to access the ServiceNow platform and perform various operations.

Import Sets
Import Sets allow users to upload data into ServiceNow from external files. The data is temporarily stored in an Import Set Table before being transformed.

Transform Maps
Transform Maps are used to map data fields from the source table to the target table.

6. SYSTEM ARCHITECTURE:

The system architecture describes how data flows from the external file into the ServiceNow database.

The process begins with an employee data file that contains employee information. This file is uploaded into the ServiceNow system using the Load Data option.

Once the file is uploaded, ServiceNow automatically creates an Import Set Table where the data is stored temporarily. This table

allows users to verify the data before it is transferred to the final table.

Next, a Transform Map is created to define the relationship between the fields in the Import Set Table and the fields in the target table.

After the mapping process is completed, the data is transformed and transferred to the target table in the ServiceNow database.

This architecture ensures that the data is organized, validated, and stored correctl.

7. MILESTONE 1 – LOAD DATA:

Milestone 1 focuses on uploading employee data into the ServiceNow system.The first step in this process is to navigate to the System Import Sets module in ServiceNow. From there, the Load Data option is selected. The employee data file stored in Excel or CSV format is then uploaded into the system.

After uploading the file, ServiceNow automatically creates an Import Set Table. This table temporarily stores the uploaded data and allows administrators to review the records.The Import Set Table plays an important role because it acts as an intermediate stage between the source file and the target table. This ensures that

the data is properly prepared before the transformation process begins.

By completing this milestone, the employee data is successfully uploaded into the ServiceNow platform.

8.MILESTONE 2 – CREATE TRANSFORM MAP:

Milestone 2 involves creating a Transform Map to transfer data from the Import Set Table to the target table.In this step, a new Transform Map is created by selecting the source table and the target table. The source table is the Import Set Table created during the data upload process, and the target table is the table where the employee data will be stored permanently.

Field mapping is then performed to match the fields from the source table with the corresponding fields in the target table. This step ensures that the data is placed in the correct fields when it is transferred.

Once the mapping is completed, the Transform Map is saved and the transformation process is executed. The data is then transferred from the Import Set Table to the target table.

9. MILESTONE 3 – DICTIONARY OVERRIDE:

Milestone 3 demonstrates the use of Dictionary Override in ServiceNow.

Dictionary Override allows administrators to modify the behavior of specific fields in a table without affecting the original configuration. This feature is useful when organizations want to customize field properties for specific tables.

For example, the Priority field in the Incident table can be modified using Dictionary Override. Administrators can change properties such as default value, visibility, or behavior of the field.This feature provides flexibility and allows organizations to customize the system according to their requirements.

10.ADVANTAGES OF AUTOMATING EMPLOYEE DATA IMPORT:

Automation of employee data import provides several advantages to organizations. One of the major benefits is the reduction of manual work. When employee data is entered manually, employees must spend a lot of time typing information into the system. Automation eliminates this problem by allowing bulk data to be uploaded directly.

Another advantage is improved data accuracy. Manual data entry often leads to typing mistakes or incorrect information being stored in the system. Automated systems ensure that data from the source file is transferred accurately into the database.

Automation also improves efficiency within the organization. Instead of spending hours entering employee records, administrators can upload the entire dataset in just a few minutes.

Another benefit is better data management. ServiceNow allows organizations to organize employee data in a structured format. This makes it easier to retrieve and update records whenever necessary.

Automation also helps organizations maintain consistency in their data. When multiple users enter data manually, different formats may be used. Automation ensures that all records follow the same structure.

11.LIMITATIONS OF THE SYSTEM:

Although automation provides many advantages, there are certain limitations that must be considered.

One limitation is the dependency on correct source data. If the source file contains incorrect or incomplete information, the system may import inaccurate data into the database.

Another limitation is the requirement for proper field mapping. If the fields in the Transform Map are not mapped correctly, the data may not appear in the correct columns of the target table.

System configuration also plays an important role. Administrators must configure Import Sets and Transform Maps carefully to ensure that the system works properly.

Additionally, users must have proper knowledge of the ServiceNow platform in order to configure the system correctly.

Despite these limitations, the benefits of automation outweigh the challenges.

12.FUTURE ENHANCEMENTS:

The project can be further improved by adding additional features. One possible enhancement is integrating the system with other enterprise applications so that employee data can be synchronized automatically.

Another improvement could be the implementation of scheduled data imports. This would allow the system to automatically import updated employee data at regular intervals without manual intervention.

The system could also include validation mechanisms that check the data before it is imported into the system. This would help prevent incorrect or incomplete data from being stored in the database.

Advanced reporting features could also be added so that administrators can generate reports related to employee data imports.

In the future, the system could be extended to support real-time data synchronization between multiple systems.

13.APPLICATIONS OF THE PROJECT:

The automated employee data import system has many practical applications in organizations.

Human Resource departments can use this system to maintain employee records efficiently. Instead of manually entering employee details, HR administrators can upload data files directly into the system.IT departments can also benefit from this system by maintaining user records in a centralized platform.

Educational institutions can use similar automation systems to manage student records and staff information.Large organizations with thousands of employees can use this system to maintain accurate and up-to-date records.

Automation helps organizations improve productivity and reduce operational costs.

14.IMPORTANCE OF SERVICENOW IN MODERN ORGANIZATIONS:

ServiceNow has become one of the most popular platforms for enterprise workflow automation. Organizations use ServiceNow to manage IT services, automate business processes, and improve operational efficiency.

One of the key advantages of ServiceNow is its ability to integrate with other systems and applications. This allows organizations to centralize their operations within a single platform.

ServiceNow also provides powerful tools for managing data and workflows. Features such as Import Sets, Transform Maps, and Dictionary Overrides allow administrators to customize the system according to organizational requirements.

By using ServiceNow, organizations can streamline their operations and improve overall productivity.

15.LITERATURE SURVEY:

from source files to the system database. Many organizations use Import Sets and TransfA literature survey is an important part of any project because it helps in understanding previous research and developments related to the topic. Many researchers and organizations have studied automation systems and data management solutions to improve organizational efficiency.

Automation of data import processes has become very common in modern organizations. Various enterprise platforms provide tools that allow data to be transferred

automatically from external systems into internal databases. ServiceNow is one such platform that offers powerful features for automation and workflow management.

Previous studies show that manual data entry often results in errors and inconsistencies. Automation helps reduce these problems by ensuring that data is transferred accurately orm Maps in ServiceNow to automate data management tasks.

The literature survey also highlights the importance of cloud-based platforms in modern enterprises. Cloud platforms allow organizations to store and manage data securely while providing easy access to users.

Through this project, the concepts discussed in previous studies are applied practically using the ServiceNow platform.

16.METHODOLOGY:

The methodology of this project explains the steps followed to implement the automated employee data import system in ServiceNow.

The first step in the methodology is preparing the employee data file. The employee information is stored in an Excel or CSV file. This file contains important details such as employee name, department, email address, and other relevant information.

The second step is uploading the data file into the ServiceNow system. This is done using the Load Data option available in the System Import Sets module.

Once the data file is uploaded, ServiceNow automatically creates an Import Set Table. This table temporarily stores the uploaded data.

The next step is creating a Transform Map. The Transform Map defines how the fields in the Import Set Table correspond to the fields in the target table.After defining the field mappings, the transformation process is executed. The data is transferred from the Import Set Table to the target table.

Finally, the results are verified to ensure that the employee data has been imported successfully.

17.DATA FLOW DESCRIPTION:

The data flow of the system explains how information moves from the source file to the ServiceNow database.The process begins with the employee data file that contains all the necessary employee details. This file acts as the input to the system.

The file is uploaded into ServiceNow using the Import Sets feature. The uploaded data is stored in the Import Set Table.

The Transform Map is then used to map the fields from the Import Set Table to the target table.During the transformation process, the system validates the data and ensures that it is placed in the correct fields.

Once the transformation process is completed, the data is stored in the final table within the ServiceNow database.This data flow ensures that the information is processed efficiently and accurately.

18.TESTING AND VALIDATION:

Testing is an important step in any system development process. The purpose of testing is to ensure that the system works correctly and produces the expected results.

In this project, testing was performed after uploading the employee data into the Import Set Table. The records were verified to ensure that the data was correctly uploaded.The Transform Map was then executed and the data was transferred to the target table. The records in the target table were checked to confirm that the field mapping was correct.

Additional validation was performed to ensure that the system handled the data correctly and that no records were lost during the transformation process.Testing confirmed that the automated employee data import system was functioning correctly.

19.PROJECT OUTCOME:

The outcome of this project demonstrates the successful implementation of an automated employee

data import system using the ServiceNow platform.The project shows that automation can significantly reduce the time required to manage employee data. Instead of manually entering records, administrators can upload entire data files and transfer them into the system automatically.

The project also highlights the importance of proper field mapping and system configuration.Through this project, students gained practical knowledge about ServiceNow features such as Import Sets, Transform Maps, and Dictionary Overrides.

The project outcome proves that automation can improve efficiency, accuracy, and productivity in organizational data management processes.

20.SUMMARY OF THE PROJECT:

The project titled Automating Employee Data Import in ServiceNow focuses on implementing an automated solution for importing employee data into the ServiceNow platform.

The project begins with preparing the employee data file in Excel or CSV format. This file contains information such as employee names, email addresses, departments, and other details.The file is then uploaded into the ServiceNow system using the Load Data option. Once the file is uploaded, an Import Set Table is created to store the data temporarily.

A Transform Map is created to define the mapping between the source table and the target table. After the mapping is completed, the data is transformed and inserted into the target table.

The project demonstrates how automation can improve efficiency and reduce manual effort in managing employee data.

21. RESULT:

The implementation of the automated employee data import process was successful. Employee data stored in the external file was uploaded into the ServiceNow system using Import Sets.

The uploaded data was stored in the Import Set Table and then transferred to the target table using Transform Maps. Field mapping ensured that all data was correctly inserted into the appropriate fields.

The Dictionary Override feature was also implemented successfully to customize field behavior.

The final result shows that the employee data import process was automated successfully, reducing manual work and improving efficiency.

22.CONCLUSION:

This project successfully demonstrated how employee data can be imported automatically into the ServiceNow platform using Import Sets and Transform Maps.

Automation helps organizations reduce manual work, improve data accuracy, and manage large volumes of employee information efficiently. The project also provided practical experience in using ServiceNow features such as field mapping and Dictionary Override.

By implementing automation, organizations can streamline their data management processes and improve productivity.Overall, ServiceNow provides a powerful platform for workflow automation and efficient data management.